

Q1

Explain the layered architecture of networks. Compare the OSI Reference Model (all 7 layers) and TCP/IP architecture (4 layers). Discuss the functions of each layer. What are the advantages and disadvantages of wired vs wireless networks?

Introduction: Computer networks use a layered architecture to break the complex problem of data communication into manageable, independent modules. Each layer performs specific functions and communicates only with the layers directly above and below it. The two dominant models are the OSI Reference Model and the TCP/IP Model.

Layered Architecture — Why? Layering reduces complexity, allows independent development of each layer, makes troubleshooting easier, and enables interoperability between different vendors' equipment. Changes in one layer don't affect others as long as interfaces remain the same.

OSI Reference Model (7 Layers):

Physical Layer (Layer 1): Transmits raw bits over a communication channel. Defines electrical, mechanical, and timing specifications. Deals with voltage levels, cable types, pin layouts, and data rates. Example: Ethernet cable, fiber optic.

Data Link Layer (Layer 2): Provides error-free transfer of frames between two directly connected nodes. Handles framing, MAC addressing, error detection (CRC), and flow control. Divided into LLC (Logical Link Control) and MAC sub-layers. Example: Ethernet, Wi-Fi (802.11).

Network Layer (Layer 3): Handles logical addressing (IP addresses) and routing — determining the best path for packets across multiple networks. Deals with packet fragmentation and reassembly. Example: IP, ICMP, routers.

Transport Layer (Layer 4): Provides end-to-end reliable (TCP) or unreliable (UDP) data delivery between processes. Handles segmentation, reassembly, flow control, congestion control, and error recovery. Port numbers identify applications.

Session Layer (Layer 5): Manages sessions (connections) between applications — establishing, maintaining, and terminating dialogues. Handles synchronization and checkpointing for long transfers.

Presentation Layer (Layer 6): Translates data between application format and network format. Handles encryption/decryption, compression, and character encoding (ASCII, Unicode). Example: SSL/TLS encryption.

Application Layer (Layer 7): Closest to the user. Provides network services to applications. Example protocols: HTTP, FTP, SMTP, DNS, SNMP.

TCP/IP Architecture (4 Layers):

Network Access Layer: Combines OSI Physical + Data Link. Handles hardware addressing (MAC), framing, and physical transmission. Protocols: Ethernet, ARP, PPP.

Internet Layer: Corresponds to OSI Network layer. Handles IP addressing and routing. Protocols: IP, ICMP, IGMP.

Transport Layer: Same as OSI Transport. TCP (reliable, connection-oriented) and UDP (unreliable, connectionless).

Application Layer: Combines OSI Session + Presentation + Application. HTTP, FTP, DNS, SMTP, SNMP.

OSI vs TCP/IP: OSI has 7 layers vs TCP/IP's 4. OSI is a theoretical reference model; TCP/IP evolved from working protocols. OSI's Session and Presentation layers are merged into TCP/IP's Application layer. TCP/IP is the model actually used on the Internet.

Wired vs Wireless Networks:

Wired (Ethernet, Fiber): High speed (up to 100 Gbps), low latency, highly secure (physical access needed to intercept), reliable, immune to interference. Disadvantage: expensive cabling, limited mobility, difficult to install in old buildings.

Wireless (Wi-Fi, 4G, 5G): Easy installation, supports mobility, flexible, cost-effective for coverage of large areas. Disadvantage: susceptible to interference (walls, microwaves, other devices), lower security (signals broadcast openly), higher latency, shared bandwidth among users, limited range.

Conclusion: The layered approach — whether OSI's 7-layer model or TCP/IP's practical 4-layer implementation — is fundamental to modern networking. Each layer's abstraction allows the Internet to function across billions of diverse devices. Understanding when to deploy wired versus wireless depends on the trade-off between mobility, cost, security, and performance requirements.

Q2

Explain different network topologies (Bus, Star, Ring, Mesh, Tree, Hybrid). Explain different transmission media (twisted pair, coaxial, fibre optic, wireless). Differentiate circuit switching vs packet switching.

Introduction: The physical and logical arrangement of a network — its topology — determines how devices communicate, how failures are handled, and how efficiently bandwidth is used. Physical layer transmission is subject to several impairments that degrade signal quality. The choice of transmission medium and switching technique further defines network performance.

Network Topologies:

Bus Topology: All devices share a single backbone cable. Simple and cheap to install. Disadvantage: a single cable failure brings down the entire network; performance degrades as more devices are added. Example: early Ethernet (10Base2).

Star Topology: All devices connect to a central hub or switch. Most common today. Failure of one device doesn't affect others; easy to add/remove devices; centralized management. Disadvantage: hub/switch failure disables entire network. Example: modern Ethernet LAN.

Ring Topology: Devices connected in a closed loop; data travels in one direction (or both in dual-ring). Deterministic access via token passing. Disadvantage: single break disrupts the ring. Example: Token Ring (IEEE 802.5), SONET.

Mesh Topology: Every device connects to every other device. Highly fault-tolerant and redundant. Disadvantage: expensive — $n(n-1)/2$ links needed for full mesh. Used in WANs and backbone networks. Example: Internet backbone.

Tree (Hierarchical) Topology: Combines star topologies in a hierarchy. Scalable and easy to manage. Used in large organizations with departments. Failure of a parent node isolates its subtree.

Hybrid Topology: Combination of two or more topologies. Most real networks are hybrid.

Transmission Impairments:

Attenuation: Loss of signal energy as it travels through the medium. Signal becomes weaker with distance. Solution: use amplifiers (analog) or repeaters (digital). Measured in decibels (dB).

Distortion: Change in signal shape, caused by different frequency components traveling at different speeds (dispersion). Particularly problematic in wired media. Solution: equalization filters.

Noise: Unwanted energy from external sources corrupting the signal. Types: thermal noise (random electron motion), intermodulation noise (signals mixing), crosstalk (signal from adjacent wire leaking in), impulse noise (sudden spikes from lightning or switching). Signal-to-Noise Ratio (SNR) measures how much signal dominates noise. Shannon's capacity theorem: $C = B \times \log_2(1 + \text{SNR})$ gives theoretical maximum channel capacity.

Transmission Media:

Twisted Pair: Two insulated copper wires twisted together to reduce crosstalk. UTP (Unshielded) is cheap and widely used for telephone and LAN (Cat5e: 1 Gbps, Cat6: 10 Gbps). STP (Shielded) adds metal shield for better noise immunity. Easy to install, low cost, limited bandwidth and distance.

Coaxial Cable: Central copper conductor surrounded by insulation, metallic shield, and outer jacket. Higher bandwidth and noise immunity than twisted pair. Used in cable TV, early Ethernet. Stiff, harder to install.

Fibre Optic: Transmits data as pulses of light through glass/plastic fibre. Extremely high bandwidth (Tbps), immune to electromagnetic interference, very low attenuation, secure (hard to tap). Expensive to install, requires skilled splicing. Single-mode fibre for long distances; multimode for shorter.

Wireless: Radio waves (Wi-Fi, Bluetooth, cellular), microwaves (satellite, point-to-point links), infrared (short range, line-of-sight). No physical cabling; supports mobility. Subject to interference and security concerns.

Circuit Switching vs Packet Switching:

Circuit Switching: A dedicated physical path is established before communication begins and held for the entire duration (e.g., traditional telephone networks). Guaranteed bandwidth, constant delay. Wasteful when channel is idle; call setup delay.

Packet Switching: Data is broken into packets that are individually routed through the network. No dedicated path; resources shared among many users. More efficient use of bandwidth, supports multiple simultaneous users, fault-tolerant. Variable delay (jitter). Used by the Internet. Packet switching advantages over circuit switching: better resource utilization (statistical multiplexing), no call setup delay, resilient to node failures (packets reroute), handles bursty traffic efficiently.

Q3

Explain error detection and correction in digital transmission: Parity Check, Checksum, CRC (Cyclic Redundancy Check) with polynomial method, and Hamming Code. What are flow control protocols? Explain Stop-and-Wait, Go-Back-N ARQ and Selective Repeat ARQ with sliding window protocol.

Introduction: In digital transmission, bits can be corrupted by noise, attenuation, or interference. Error control mechanisms detect and correct these errors to ensure reliable data delivery. Flow control prevents a fast sender from overwhelming a slow receiver. ARQ (Automatic Repeat reQuest) protocols combine both.

Error Detection Techniques:

Parity Check: A single parity bit is appended to each data unit. Even parity: total number of 1s (including parity bit) is even. Odd parity: total is odd. Simple, detects single-bit errors. Cannot detect even numbers of errors or correct errors. Example: data 1010001 → even parity bit = 1 → transmitted: 10100011.

Checksum: Data is divided into fixed-size segments. All segments are added together (1's complement arithmetic). The result is complemented and appended as the checksum. Receiver recomputes; if all-ones result → no error. Used in TCP, UDP, IP headers. Better than parity, detects most burst errors.

CRC — Cyclic Redundancy Check: Most powerful error detection. Treats the data as a large binary polynomial (message polynomial $M(x)$) and divides it by a generator polynomial $G(x)$. The remainder r is appended to the message. Receiver divides the received message by $G(x)$; if remainder is 0 → no error. Detects all single-bit errors, all double-bit errors, all odd numbers of errors, all burst errors of length $\leq$ degree of $G(x)$. Example: data = 1101011011, generator = 10011 → CRC computed by binary long division.

Error Correction — Hamming Code: Adds redundancy bits at power-of-2 positions. For m data bits, r parity bits needed where $2^r \geq m + r + 1$. Each parity bit covers specific bit positions. On receiving, recalculate all parity bits — the binary pattern of error positions (syndrome) directly gives the bit position that is wrong. Example: 4 data bits need 3 parity bits (positions 1, 2, 4). Can correct single-bit errors, detect double-bit errors.

Flow Control Protocols:

Stop-and-Wait ARQ: Sender transmits one frame, then waits for acknowledgement (ACK) before sending the next. Simple but highly inefficient — sender idle during propagation delay. Efficiency = $1/(1 + 2a)$ where a = propagation delay/transmission time.

Go-Back-N ARQ: Sender can have up to W (window size) unacknowledged frames in transit. If frame n is lost or corrupted, receiver discards frames $n, n+1, n+2, \dots$ and sender retransmits ALL from frame n onwards. Window size $\leq 2^n - 1$ (n = sequence number bits). Wastes bandwidth on retransmitting correct frames.

Selective Repeat ARQ: Like Go-Back-N but receiver buffers out-of-order frames. Only the specific corrupted frame is retransmitted. More efficient than Go-Back-N but requires larger receiver buffer. Window size $\leq 2^n - 1$.

Sliding Window Concept: Maintains two windows — sender window (frames that can be sent without ACK) and receiver window (frames that can be accepted). Window slides forward as ACKs are received. Allows pipelining — multiple frames in transit simultaneously, keeping the channel busy. **Conclusion:** Error detection techniques from simple parity to powerful CRC form the first defence against transmission errors, while Hamming code provides self-correction capability. ARQ protocols — from the simple but inefficient Stop-and-Wait to the bandwidth-efficient Selective Repeat — build reliable data links over unreliable physical channels using the sliding window principle.

Q4

Explain MAC (Medium Access Control) protocols: Pure ALOHA and Slotted ALOHA (with efficiency calculations), CSMA/CD — p-persistent, 1-persistent, non-persistent. Also explain functions of Switches and Routers and at which OSI layer they operate.

Introduction: In shared communication channels — such as a wireless network or early Ethernet bus — multiple devices compete for the medium simultaneously. Medium Access Control (MAC) protocols govern who can transmit when, preventing or recovering from collisions. This answer covers ALOHA, CSMA/CD variants, and the network devices that operate at the data link and network layers.

Pure ALOHA: Developed at the University of Hawaii (1970). Any station can transmit at any time. If a collision occurs (two stations transmit simultaneously), both wait a random backoff time and retransmit. Collision detected by lack of ACK within timeout. Maximum throughput: $S = G \times e^{-(2G)}$, peaks at $G = 0.5$, giving maximum efficiency of $1/(2e) \approx 18.4\%$. Very simple but highly inefficient — vulnerable time is $2 \times$ frame transmission time.

Slotted ALOHA: Time is divided into equal slots (= one frame transmission time). Stations can only begin transmission at the start of a slot, halving collision probability. Maximum throughput: $S = G \times e^{-(G)}$, peaks at $G = 1$, giving efficiency of $1/e \approx 36.8\%$ — double that of Pure ALOHA. Example: if $G = 1$ (one frame per slot on average), $P(\text{success}) = e^{-(1)} \approx 0.368$. Used in satellite communications and early GSM.

CSMA — Carrier Sense Multiple Access: Before transmitting, a station listens to the channel (carrier sense). If busy, it waits. If idle, it transmits. Reduces collisions but doesn't eliminate them (propagation delay means two stations may sense idle simultaneously).

1-Persistent CSMA: If channel busy, wait and transmit as soon as it becomes idle (probability 1). Greedy — high collision probability when multiple stations waiting.
Non-Persistent CSMA: If channel busy, wait a random time then sense again. Less greedy, better channel utilization, but longer delays when traffic is low.
p-Persistent CSMA: If channel idle, transmit with probability p ; defer to next slot with probability $(1-p)$. Balances between 1-persistent and non-persistent.

CSMA/CD (Collision Detection): Used in Ethernet (IEEE 802.3). Station senses the channel; if idle, transmits. While transmitting, continues to listen. If collision detected (signal level doubles), immediately stops, sends jam signal (32 bits) to ensure all stations detect the collision, then waits a random backoff time before retrying. Binary Exponential Backoff: after k -th collision, wait random time from $\{0, 1, ..., 2^k - 1\}$ slot times. Minimum frame size = $2 \times$ propagation delay $\times$ bandwidth (ensures frame still on wire when collision detected). CSMA/CA (Collision Avoidance) is used in wireless (Wi-Fi) since detecting collisions on wireless is impractical.

Switches vs Routers:

Switch (Layer 2 — Data Link): Connects devices within the same LAN. Forwards frames based on MAC addresses. Maintains a MAC address table (CAM table) — maps MAC addresses to physical ports. Learns by observing source MAC of incoming frames. Reduces collision domains (each port is its own domain). Supports full-duplex. Does not cross network boundaries. Example: 24-port office switch.
Router (Layer 3 — Network): Connects different networks. Forwards packets based on IP addresses. Maintains routing tables populated by static configuration or dynamic routing protocols (RIP, OSPF, BGP). Performs NAT (Network Address Translation), firewall functions, and QoS. Separates broadcast domains. Example: home broadband router, enterprise core router.

Conclusion: MAC protocols evolved from the simple but inefficient Pure ALOHA to CSMA/CD which dramatically improves channel utilization through carrier sensing and collision detection. Switches and routers operate at different OSI layers and serve complementary roles — switches organize traffic within a network segment efficiently using MAC addressing, while routers intelligently direct traffic between networks using IP addressing and routing algorithms.

Q5

Explain IP addressing — IPv4 classes (A, B, C, D, E), subnetting, CIDR. Compare IPv4 and IPv6 (why IPv6 is better). Explain DHCP and ARP. Describe private IP address ranges and why they are needed. Sketch the IP datagram format and explain each field.

Introduction: The Network layer is responsible for logical addressing and routing — determining how packets travel from source to destination across interconnected networks. IP (Internet Protocol) is the foundation of this layer, providing the addressing scheme. DHCP automates address assignment, while ARP resolves IP addresses to physical MAC addresses.

IPv4 Address Classes: IPv4 addresses are 32-bit numbers in dotted-decimal notation (e.g., 192.168.1.1). Divided into five classes:
Class A: First bit = 0. Range 1–126.x.x.x. Network: 8 bits, Host: 24 bits. 126 networks, ~16.7 million hosts each. For large organizations. Private: 10.x.x.x.
Class B: First bits = 10. Range 128–191.x.x.x. Network: 16 bits, Host: 16 bits. 16,384 networks, 65,534 hosts each. Medium organizations. Private: 172.16–31.x.x.
Class C: First bits = 110. Range 192–223.x.x.x. Network: 24 bits, Host: 8 bits. ~2 million networks, 254 hosts each. Small organizations. Private: 192.168.x.x.
Class D (224–239.x.x.x): Multicast. Class E (240–255.x.x.x): Reserved/experimental.
Special: 127.0.0.1 = loopback. 255.255.255.255 = broadcast.

Subnetting & CIDR: Subnetting divides a network into smaller subnets using a subnet mask. Example: 192.168.1.0/26 gives 4 subnets with 62 usable hosts each ($64 - 2$ for network and broadcast addresses). CIDR (Classless Inter-Domain Routing) uses prefix notation (/n) and eliminates rigid class boundaries, reducing routing table size through route aggregation (supernetting).

Private IP Addresses & NAT: Private ranges (10.x.x.x, 172.16–31.x.x, 192.168.x.x) are non-routable on the Internet, used within private networks. NAT (Network Address Translation) maps multiple private IPs to one public IP at the router, conserving IPv4 addresses. Advantage of subnetting: improves security, reduces broadcast domain size, better network management.

IPv4 Datagram Format Fields: Version (4 bits), Header Length (4 bits), Type of Service (8 bits, QoS), Total Length (16 bits), Identification + Flags + Fragment Offset (fragmentation), TTL (Time to Live — decremented each hop; prevents infinite loops), Protocol (TCP=6, UDP=17, ICMP=1), Header Checksum, Source IP, Destination IP, Options.

IPv6 — Why Better than IPv4: IPv6 uses 128-bit addresses ($2^{128} \approx 3.4 \times 10^{38}$ — effectively unlimited). Written as eight groups of hex digits. Key improvements: no address exhaustion, no NAT needed, built-in IPsec security, simpler fixed-size header (40 bytes, no checksum — left to transport layer), stateless address auto-configuration (SLAAC — no DHCP needed), better support for QoS and multicast, no broadcast (uses multicast instead), more efficient routing.

DHCP — Dynamic Host Configuration Protocol:

Automatically assigns IP addresses to hosts. Process (DORA):
1. *Discover:* Client broadcasts DHCP Discover (no IP yet).
2. *Offer:* DHCP server responds with IP address offer, subnet mask, gateway, DNS, lease time.
3. *Request:* Client broadcasts which offer it accepts.
4. *Acknowledge:* Server confirms the lease.

Lease is temporary — client must renew before expiry. DHCP removes manual IP configuration, prevents conflicts, and simplifies management.

ARP — Address Resolution Protocol: Bridges the gap between Layer 3 (IP) and Layer 2 (MAC). When a device wants to send a packet to an IP address on the same network, it doesn't know the MAC address. ARP broadcasts: "Who has IP 192.168.1.5? Tell 192.168.1.1." The device with that IP replies with its MAC address. The mapping is cached in the ARP table for future use. Gratuitous ARP: a device announces its own IP-to-MAC mapping (used to detect IP conflicts and update caches after NIC replacement).

Conclusion: IP addressing forms the logical foundation of Internet communication. IPv4's class-based scheme, while elegantly simple, has given way to CIDR and eventually IPv6 to meet the demands of a connected world. DHCP automates the critical task of address configuration, while ARP seamlessly bridges the logical IP world with the physical MAC addressing of the data link layer.

Q6

Explain routing protocols and algorithms: Distance Vector Routing (Bellman-Ford, count-to-infinity problem and 3 solutions), Link State Routing (Dijkstra's algorithm, OSPF). Explain ICMP — its purpose and message types. Explain unicast vs multicast routing.

Introduction: Routing is the process of selecting paths in a network along which to send data packets. Routing algorithms determine these paths. Static routing uses manually configured tables; dynamic routing uses protocols that automatically adapt to topology changes. This answer covers the two major dynamic routing paradigms and ICMP — the network layer's error and diagnostic protocol.

Distance Vector Routing (Bellman-Ford Algorithm): Each router maintains a distance vector table containing the best-known distance (hop count or metric) to every other router and which neighbour to go through. Periodically, each router sends its entire distance vector to its directly connected neighbours. Neighbours update their tables based on received vectors: $D(x, y) = \min \text{over all neighbours } v \{ \text{cost}(x, v) + D(v, y) \}$.

Convergence is gradual — good news (new short path) spreads fast, bad news (link failure) spreads slowly.
Count-to-Infinity Problem: When a link fails, routers may keep incrementing the distance to unreachable nodes, bouncing the metric up indefinitely. Example: A–B–C, if A–B link fails, B thinks it can reach A via C (C had A in its table from before), C thinks it can reach A via B → they loop, incrementing to infinity.

Solutions:

Split Horizon: A router never advertises a route back through the interface it learned it from. Prevents two-node loops.

Poison Reverse: Advertise failed routes with infinite metric (16 in RIP) back to the neighbour. More aggressive than split horizon.

Hold-Down Timers: When a route goes down, ignore any updates about that route for a fixed period — prevents premature re-addition of a bad route. Protocol: RIP (Routing Information Protocol) uses distance vector, max 15 hops, updates every 30 seconds.

Link State Routing (Dijkstra's Algorithm): Each router builds a complete map of the network topology. Process: Each router discovers its neighbours (Hello packets) and measures link costs.

Each router broadcasts Link State Advertisements (LSAs) to ALL routers in the network (flooding).

Every router builds identical network topology graph. Each router runs Dijkstra's Shortest Path First (SPF) algorithm on its map to compute best paths.

Dijkstra's: Start with source, tentatively assign infinite distance to all, zero to source. Repeatedly pick the unvisited node with smallest tentative distance, update its neighbours. Converges to shortest paths in $O((V+E) \log V)$. Advantages over distance vector: Faster convergence, no count-to-infinity, scales better with OSPF hierarchical areas. Disadvantage: more memory and CPU (each router stores full topology).

Protocol: OSPF (Open Shortest Path First) — link state, supports VLSM/CIDR, areas reduce flooding scope, uses cost = $10^8/\text{bandwidth}$ as metric.

ICMP — Internet Control Message Protocol: ICMP is used by network devices to send error messages and operational information. It rides inside IP packets (Protocol = 1). Key message types:

Echo Request/Reply (Type 8/0): Used by the ping command to test reachability. Sender sends Echo Request to destination IP; if reachable, destination replies with Echo Reply. Measures RTT.

Destination Unreachable (Type 3): Router or host cannot deliver packet. Codes indicate reason: Network unreachable (0), Host unreachable (1), Port unreachable (3), Fragmentation needed but DF bit set (4).

Time Exceeded (Type 11): TTL reached 0 before packet reached destination (Code 0 — used by traceroute). Each hop decrements TTL; when it hits 0 the router sends Time Exceeded back to source. Traceroute sends packets with TTL=1, 2, 3... to map each router hop.

Redirect (Type 5): Router tells host to use a better route.
Source Quench (Type 4, deprecated): Congestion control signal asking sender to slow down.

Unicast vs Multicast Routing: Unicast: one sender to one specific receiver. Standard IP forwarding. Multicast: one sender to a group of interested receivers. Uses Class D addresses (224.0.0.0–239.255.255.255). IGMP (Internet Group Management Protocol) lets hosts join multicast groups. PIM (Protocol Independent Multicast) routes multicast traffic efficiently, building a distribution tree. Much

Q7

Discuss the distinguishing features of TCP in detail: three-way handshake for connection setup (why not two-way?), reliable data transfer, flow control (sliding window), congestion control (slow start, congestion avoidance, fast retransmit), RTT estimation, and retransmissions. Also explain TCP connection management finite state machine.

Introduction: TCP (Transmission Control Protocol) is the cornerstone of reliable Internet communication. While IP provides best-effort packet delivery with no guarantees, TCP builds a reliable, ordered, connection-oriented service on top of IP. Understanding TCP's mechanisms is essential to understanding how the Internet actually achieves reliability for applications like web browsing, email, and file transfer.

Key Features of TCP: Connection-oriented (logical connection established before data transfer), full-duplex (simultaneous bidirectional data flow), byte-stream service (no message boundaries), reliable delivery (acknowledgements, retransmission), ordered delivery (sequence numbers), flow control (receiver window), congestion control (adaptive sending rate), error detection (16-bit checksum).

TCP Segment Header Fields: Source Port, Destination Port, Sequence Number (byte offset of first byte in segment), Acknowledgement Number (next expected byte), Header Length, Flags (SYN, ACK, FIN, RST, PSH, URG), Receive Window (flow control), Checksum, Urgent Pointer, Options (MSS, SACK, timestamps).

Three-Way Handshake (Connection Establishment):

1. **SYN:** Client sends segment with SYN flag set, ISN (Initial Sequence Number) = x. "I want to connect, my sequence starts at x."
2. **SYN-ACK:** Server responds with SYN + ACK. Server's ISN = y, ACK = x+1. "I accept, my sequence starts at y, I expect byte x+1 from you."
3. **ACK:** Client sends ACK = y+1. "I expect byte y+1 from you." Connection established.

Why Not Two-Way Handshake? A two-way handshake would only confirm one direction of communication. The server needs to know the client received its SYN-ACK and is ready to receive. Without the third ACK, the server doesn't know if its SYN-ACK arrived, and couldn't safely establish its receive window. Also prevents old duplicate connection requests from creating spurious connections.

Reliable Data Transfer: Every segment is acknowledged. If ACK not received within timeout (RTO), the segment is retransmitted. Cumulative ACKs acknowledge all bytes up to ACK number. TCP uses piggybacking — ACKs sent with data segments in the reverse direction.

RTT Estimation: Timeout (RTO) must adapt to network conditions. TCP measures SampleRTT for each segment. $EstimatedRTT = (1-\alpha) \times EstimatedRTT + \alpha \times SampleRTT$ ($\alpha = 0.125$). DevRTT measures variability. $RTO = EstimatedRTT + 4 \times DevRTT$. If timeout occurs, RTO is doubled (exponential backoff).

Flow Control — Receiver Window: Receiver advertises its available buffer space (rwnd) in every ACK. Sender limits unacknowledged data to $\min(cwnd, rwnd)$. Prevents fast sender overwhelming slow receiver. If $rwnd = 0$, sender stops but sends 1-byte probe segments to check when window reopens.

Congestion Control: *Slow Start:* cwnd starts at 1 MSS, doubles every RTT (exponential growth) until ssthresh is reached. *Congestion Avoidance:* Above ssthresh, cwnd increases by 1 MSS per RTT (linear growth). *Fast Retransmit:* On receiving 3 duplicate ACKs, immediately retransmit (don't wait for timeout). $ssthresh = cwnd/2$, $cwnd = ssthresh + 3$. *Fast Recovery:* After fast retransmit, continue in congestion avoidance (not slow start). *On timeout:* $ssthresh = cwnd/2$, $cwnd = 1$ MSS, restart slow start.

TCP Connection Termination — Four-Way Handshake:

Client FIN → Server ACK → (server continues sending if needed) → Server FIN → Client ACK. Client waits 2×MSL (Maximum Segment Lifetime = 2 min) in TIME-WAIT before fully closing — ensures final ACK reached server.

Conclusion: TCP's elegance lies in achieving reliable, ordered, flow-controlled, and congestion-controlled communication over the unreliable IP layer. The three-way handshake establishes mutual readiness; sequence numbers and ACKs ensure reliability; the sliding window provides flow control; and the AIMD (Additive Increase Multiplicative Decrease) congestion control algorithm ensures TCP flows share the network fairly without overwhelming it.

Q8

Explain network security: user authentication, data integrity, and cryptography (symmetric — DES, AES; asymmetric — RSA). Explain Firewalls — types and configurations (packet filtering, stateful, proxy). Explain DNS working and chain of DNS servers. Explain Quality of Service (QoS) parameters and how to improve QoS.

Network Security: Cryptography, Authentication, Firewalls, DNS & QoS

Introduction: As networks carry increasingly sensitive data, security becomes paramount. Network security encompasses mechanisms to ensure confidentiality (data not readable by unauthorised parties), integrity (data not altered in transit), authentication (verifying identity), and availability (services remain accessible). This answer covers cryptographic algorithms, authentication, firewalls, DNS, and Quality of Service.

Authentication: Authentication verifies that a communicating entity is who it claims to be. Methods: *Password-based:* Username + password. Simple but vulnerable to brute force, dictionary attacks, and eavesdropping if transmitted in plain text. *Challenge-Response:* Server sends a random challenge; client must respond with a value computed using a shared secret (e.g., HMAC-SHA256 of challenge+secret). Proves knowledge of secret without transmitting it. *Digital Certificates (PKI):* A Certificate Authority (CA) signs a certificate binding a public key to an identity. Used in HTTPS — browser verifies the server's certificate was signed by a trusted CA. *Multi-Factor Authentication (MFA):* Combines something you know (password), something you have (OTP on phone), and something you are (fingerprint).

Data Integrity: Achieved using cryptographic hash functions (MD5 — broken, SHA-1 — deprecated, SHA-256 — current standard). A hash is a fixed-size fingerprint of data. Any change to data produces a completely different hash (avalanche effect). To ensure integrity: sender computes hash, sends data + hash. Receiver recomputes hash; if it matches → data unmodified. HMAC (Hash-based Message Authentication Code) adds a secret key to the hash for combined integrity + authentication.

Symmetric Key Cryptography: Same key for encryption and decryption. Fast — suitable for bulk data. Key distribution is the main challenge. *DES (Data Encryption Standard):* 56-bit key, 16 rounds of substitution-permutation. Now broken — brute-forceable. *3DES:* Applies DES three times with different keys. 168-bit effective security. Slow. *AES (Advanced Encryption Standard):* 128/192/256-bit keys, 10/12/14 rounds. Current gold standard. Hardware-accelerated in modern CPUs. Used in HTTPS, disk encryption, VPNs.

Asymmetric Key Cryptography: Public key + private key pair. Encrypt with public key → only private key decrypts. Solves key distribution problem. *RSA:* Based on difficulty of factoring large primes. 2048-bit keys typical. Slow — used for key exchange and digital signatures, not bulk data. Example: HTTPS uses RSA to exchange a symmetric session key.

Firewalls:

Packet Filtering (Stateless): Examines each packet's source/destination IP, port, and protocol against ACL rules. Fast but cannot detect attacks spanning multiple packets. *Stateful Inspection:* Tracks TCP connection state. Allows inbound packets only for established outbound sessions. Much more secure than stateless.

Application-Layer Gateway (Proxy Firewall): Terminates the connection, inspects application content, then opens new connection if legitimate. Can filter URLs, commands, and content. Slowest but most thorough.

DNS — Domain Name System: DNS translates domain names to IP addresses. Hierarchy: Root servers → TLD servers (.com, .in) → Authoritative name servers. Resolution: Recursive resolver queries root → TLD → authoritative → returns IP to client. DNS records: A (IPv4), AAAA (IPv6), MX (mail), CNAME (alias), NS (name server). DNS uses UDP port 53 for queries, TCP port 53 for zone transfers.

Quality of Service (QoS): QoS guarantees performance levels for specific traffic types. Parameters: Bandwidth (throughput), Latency (delay), Jitter (variation in delay), Packet loss. Critical for real-time applications (VoIP, video conferencing) that cannot tolerate high latency or jitter. Techniques to improve QoS: *Traffic Classification:* Identify and mark packets by type (voice, video, data). *Prioritisation (QoS Queuing):* High-priority traffic (VoIP) served before bulk traffic (downloads). Priority Queuing, Weighted Fair Queuing (WFQ). *Traffic Shaping:* Leaky Bucket — enforces constant output rate regardless of burst input. Token Bucket — allows bursts up to a token bucket size while enforcing.

Q1 Distinguish between Internet, Intranet & Extranet. Structure of Internet, Modes of Connecting, How a Modem Works.

Introduction : The Internet, Intranet, and Extranet are three related but distinct network concepts.

Understanding their differences is fundamental to grasping modern digital communication.

•**Internet:** A global, public network of interconnected computers using TCP/IP protocols. Open to everyone — billions of users, devices, and servers worldwide share information freely.

•**Intranet:** A private network within an organization. Uses the same technologies as the Internet (HTML, HTTP, TCP/IP) but access is restricted to authorized employees. Example: A company’s internal HR portal.

•**Extranet:** An extension of an intranet that selectively allows external parties (vendors, partners, customers) to access specific internal resources. It bridges the private intranet and the public internet.

Key Differences Table

Internet: Public, global, no ownership. Intranet: Private, organizational, owned by company. Extranet: Semi-private, shared with trusted partners.

Structure of the Internet

The Internet is structured as a layered, hierarchical network:

•Tier 1 Networks: Large backbone providers (AT&T, NTT) interconnect globally via peering agreements. These form the "backbone" of the Internet.

•Tier 2 & 3 Networks: Regional and local ISPs connect to Tier 1 networks and provide Internet access to end users and businesses.

•End Devices: PCs, smartphones, servers — all connected via ISPs.

•Data travels as packets, routed across multiple networks using routers, switches, and fiber-optic or wireless links.

How the Internet Works

When you visit a website, your browser sends an HTTP request. Your ISP routes this request through multiple routers. The destination server receives the request and sends back data in packets. Packets travel independently, may take different routes, and are reassembled at the destination. This entire process relies on the TCP/IP protocol suite.

Modes of Connecting to the Internet

•Dial-Up: Uses telephone lines and a modem. Slow (max 56 kbps). Nearly obsolete today.

•DSL (Digital Subscriber Line): Uses existing telephone wiring but provides much faster speeds. ADSL (Asymmetric DSL) is most common — faster download than upload.

•Cable: Uses TV cable infrastructure. Faster than DSL. Speeds up to hundreds of Mbps.

•Fiber Optic: Uses light signals in glass/plastic fiber. Extremely high speed (Gbps range). Most modern broadband technology.

•Satellite: Useful in remote areas. Higher latency. Providers like Starlink offer low-earth-orbit satellite internet.

•Mobile (3G/4G/5G): Cellular network-based internet access. 5G offers ultra-low latency and multi-Gbps speeds.

•Wi-Fi: Wireless local area connection to a router which itself connects to the Internet via one of the above methods.

•Leased Lines: Dedicated private connections for businesses. Guaranteed bandwidth and uptime.

How a Modem Works

A modem (MOdulator-DEModulator) is a hardware device that converts digital data from a computer into analog signals suitable for transmission over telephone lines, and vice versa.

Modulation (Sending Data)

When you send data, the modem takes the binary digital data (0s and 1s) from your computer and modulates (converts) it into an analog signal — varying the frequency, amplitude, or phase of a carrier wave — so it can travel over copper telephone wires.

Demodulation (Receiving Data)

When receiving, the modem does the reverse: it demodulates the incoming analog signal, extracting the encoded digital data and passing it to the computer.

Q2 IP Addressing Scheme — Classes A, B, C, D, E. DNS, ISP, Internet Congestion, Collaborative Computing.

Introduction to IP Addressing

An IP (Internet Protocol) address is a unique numerical label assigned to every device connected to a network. It enables devices to identify and communicate with each other on the Internet or on a local network.

IP Addressing Scheme

IPv4 addresses are 32-bit numbers, written in dotted-decimal notation as four octets separated by dots.

Example: 192.168.1.10.

Each octet can range from 0 to 255, giving a theoretical maximum of over 4 billion unique addresses. The address is divided into a Network part (identifies the network) and a Host part (identifies the device within that network).

How Addresses Are Assigned

•Static Assignment: A fixed IP manually configured by an administrator. Used for servers and network devices.

•Dynamic Assignment: The DHCP (Dynamic Host Configuration Protocol) server automatically assigns an IP from a pool when a device connects. Most home and office devices use this.

•IANA (Internet Assigned Numbers Authority) oversees global allocation; IANA assigns to Regional Internet Registries (RIRs), which assign to ISPs, who assign to end users.

IP Address Classes

IPv4 addresses are grouped into 5 classes based on the leading bits of the address:

•Class A: Range 1.0.0.0 – 126.255.255.255. First bit: 0. Default mask: 255.0.0.0 (/8). Supports ~16 million hosts per network. Used for very large organizations.

•Class B: Range 128.0.0.0 – 191.255.255.255. First bits: 10. Default mask: 255.255.0.0 (/16). Supports ~65,000 hosts per network. Used for medium organizations.

•Class C: Range 192.0.0.0 – 223.255.255.255. First bits: 110. Default mask: 255.255.255.0 (/24). Supports 254 hosts per network. Most commonly used.

•Class D: Range 224.0.0.0 – 239.255.255.255. Reserved for multicast (sending data to multiple recipients simultaneously).

•Class E: Range 240.0.0.0 – 255.255.255.255. Reserved for experimental use. Not available for general assignment.

DNS (Domain Name System)

DNS is often called the "phonebook of the Internet." It translates human-readable domain names (like www.google.com) into machine-readable IP addresses (like 142.250.80.36).

•DNS Resolver: Your ISP’s server that accepts DNS queries from clients.

•Root Servers: 13 sets of root name servers hold the top-level domain information.

•TLD Servers: Handle top-level domains like .com, .org

•Authoritative Name Servers: Hold the actual DNS records for specific domains.

ISP (Internet Service Provider)

An ISP is a company that provides access to the Internet. They maintain the infrastructure (servers, routers, cables) and provide connectivity to homes and businesses in exchange for a subscription fee. ISPs operate at Tier 1, 2, or 3 levels.

Collaborative Computing

Collaborative computing refers to the use of networked computers and software to enable multiple users to work together on shared tasks — e.g., Google Docs, Microsoft 365, video conferencing tools.

Internet Congestion

Internet congestion occurs when the volume of data traffic on a network segment exceeds its capacity, causing slowdowns, packet loss, and increased latency.

•Causes: High number of simultaneous users, bandwidth-intensive applications (video streaming), insufficient infrastructure capacity, network attacks (DDoS).

•Effects: Slow page loads, video buffering, dropped packets, VoIP call degradation.

•Solutions: Traffic shaping and QoS (Quality of Service) prioritizes critical traffic. CDNs (Content Delivery Networks) like Cloudflare distribute content geographically. Load balancing distributes traffic across multiple servers. Network upgrades (fiber optic, 5G) increase capacity.

Q3 Explain FTP, HTTP, Telnet & Gopher. Role of HTTP & Web Browsers in WWW. Search Engines: Directory vs Meta.

Introduction

The World Wide Web relies on several application-layer protocols for communication between clients and servers. Understanding FTP, HTTP, Telnet, and Gopher helps explain the evolution of network services.

FTP (File Transfer Protocol)

FTP is a standard protocol for transferring files between a client and server over a TCP/IP network. It uses two channels: port 21 for commands and port 20 for data transfer.

•Active FTP: Server initiates the data connection back to the client.

•Passive FTP: Client initiates both connections (firewall-friendly).

•FTP transfers files in binary or ASCII mode. SFTP and FTPS are secure variants.

Telnet

Telnet is a protocol that provides a command-line interface to a remote computer over a network (port 23). It allows users to log in and execute commands on remote machines. •Major limitation: Telnet transmits data in plain text — including passwords — making it insecure. Replaced by SSH (Secure Shell) which encrypts all communication.

Gopher

Gopher (port 70) was an early Internet protocol for distributing, searching, and retrieving documents. It presented a hierarchical menu-based system (like a directory tree) for navigating content.

•Predated the World Wide Web. Uses a simple text-based menu structure.

•Largely obsolete; replaced by HTTP/WWW which supports richer content including multimedia.

HTTP & Web Browsers in the WWW

HTTP (HyperText Transfer Protocol) is the foundation of data communication on the Web. It operates as a request-response protocol:

•Client (browser) sends an HTTP Request to the server (e.g., GET /index.html HTTP/1.1).

•Server processes the request and sends back an HTTP Response with status code (200 OK, 404 Not Found, etc.) and the requested content.

•HTTP is stateless — each request is treated independently with no memory of prior interactions.

•HTTPS (HTTP Secure) encrypts communication using TLS/SSL, protecting data in transit.

Web browsers parse and render HTML, CSS, and JavaScript, displaying rich, multimedia web pages. Browsers also handle cookies, caching, and browser history.

Popular Web Browsers

•Google Chrome: Most widely used. Fast, secure, extensive extension library.

•Mozilla Firefox: Open-source, privacy-focused.

•Microsoft Edge: Built on Chromium, integrated with Windows.

•Apple Safari: Default browser on macOS and iOS. Optimized for Apple hardware.

•Opera: Known for built-in VPN and data-saving features.

•Brave: Privacy-first browser with built-in ad blocker.

How a Search Engine Works

A search engine has three main components:

•1. Crawling: Automated bots (spiders/crawlers) browse the web by following hyperlinks from page to page, downloading content for indexing.

•2. Indexing: The downloaded content is analyzed and stored in a massive database (the index). Pages are categorized by keywords, metadata, and relevance signals.

•3. Ranking/Retrieval: When a user enters a query, the engine searches its index and applies ranking algorithms (like Google PageRank) to return the most relevant results in order of relevance.

Directory vs Meta Search Engines

•Directory Search Engines: Maintain a human-curated hierarchy of websites organized into categories. Example: Yahoo! Directory (now discontinued), DMOZ. Pros: High quality, organized. Cons: Limited coverage, slow to update.

•Meta Search Engines: Do not maintain their own index. Instead, they send queries to multiple search engines simultaneously and aggregate/compile the results.

Examples: Dogpile, MetaCrawler. Pros: Broader coverage. Cons: Less control over result quality, slower response.

•Standard Search Engines (Web Crawlers): Maintain their own index via crawling. Examples: Google, Bing, DuckDuckGo. Most powerful and widely used.

Q4 TCP/IP Model & Core Protocols. What is HTML? Basic Structure, Important Tags, Hyperlinks & Plug-in Architecture. Introduction to TCP/IP Model

The TCP/IP (Transmission Control Protocol / Internet Protocol) model is the practical framework that governs how data is transmitted across the Internet. It consists of four layers (a simplified alternative to the 7-layer OSI model).

TCP/IP Layers & Core Protocols

- 1. Application Layer: The topmost layer, directly interacting with software applications. Protocols: HTTP/HTTPS (web), FTP (file transfer), SMTP/POP3/IMAP (email), DNS (name resolution), SSH (secure shell), Telnet.
- 2. Transport Layer: Provides end-to-end communication between applications. TCP (Transmission Control Protocol): Connection-oriented, reliable, ordered delivery with error checking. Three-way handshake (SYN, SYN-ACK, ACK). UDP (User Datagram Protocol): Connectionless, fast, no guaranteed delivery. Used for streaming, gaming.
- 3. Internet Layer: Responsible for logical addressing and routing packets across networks. Core protocol: IP (Internet Protocol). Also includes ICMP (ping/traceroute) and ARP (address resolution).
- 4. Network Access Layer (Link Layer): Handles physical transmission of data over the local network. Protocols: Ethernet, Wi-Fi (802.11), MAC addressing.

HTML — Introduction

HTML (HyperText Markup Language) is the standard markup language used to create web pages. It describes the structure and content of a web page using a series of elements represented by tags.

- HTML is not a programming language — it cannot perform computations. It describes document structure.
- An HTML document is a plain text file with special markup tags. Browsers read HTML and render the visual web page.
- Current standard: HTML5.

Basic HTML Structure

Every HTML document follows this skeleton:
<!DOCTYPE html> — declares the document type.
<html> — root element.
<head> — metadata (title, CSS links, scripts).
<title> — tab/window title.
<body> — visible content of the page.

Important HTML Tags

- <h1> to <h6>: Headings, h1 being the largest.
- <p>: Paragraph text.
- : Hyperlink (anchor tag).
- : Embed an image.
- , , : Unordered / ordered lists.
- <table>, <tr>, <td>, <th>: Tables.
- <form>, <input>, <button>: Forms for user input.
- <div>, : Generic container elements for layout/styling.

How Hyperlinks Are Created

- Hyperlinks are created using the anchor (<a>) tag with the href (Hypertext REference) attribute:
- Visit Google
 - External links: Full URL (https://...)
 - Internal links: Relative path (/about.html)
 - Email links: href="mailto:user@example.com"
 - Phone links: href="tel:+91XXXXXXXXXX"
 - Anchor links: href="#section-id" (jumps to a section on the same page)
 - target="_blank" opens the link in a new tab.

Architecture of Plug-ins

- Plug-ins (also called plugins or browser plugins) are software components that extend the functionality of a web browser beyond what HTML/CSS/JS natively provide.
- Integration: Plug-ins are integrated into the browser framework. When the browser encounters content that requires a plugin (identified by the MIME type), it loads the plugin to handle that content inline within the web page.
 - Examples: Adobe Flash Player (for animations/games), Adobe Reader (PDFs in browser), Java Applet plugin, QuickTime (media).

- Plugin API: Browsers expose a Netscape Plugin API (NPAPI) or newer PPAPI (Pepper Plugin API in Chrome) that allows plugins to interface with browser rendering.

Plug-in Architecture Details

- Discovery: On page load, browser checks MIME types of embedded content (e.g., application/pdf). If a registered plugin handles that MIME type, the plugin is invoked.
- Rendering: The plugin renders its content in a rectangular area within the browser window. The browser manages the surrounding HTML; the plugin manages its own area.
- Communication: Plugins can communicate with JavaScript via the DOM (Document Object Model), enabling dynamic interaction.
- Security Concerns: Plugins (especially Flash) became major security vulnerabilities. Most modern browsers have deprecated NPAPI plugins

Q5 Components of E-mail Architecture. How E-mail Works. SMTP & MIME — Header Fields, Types, Mailer Features & Message Composition.

Introduction

Electronic mail (e-mail) is one of the most widely used services on the Internet. Understanding e-mail architecture, protocols, and the MIME extension is essential for any IT professional.

Components of E-mail Architecture

- MUA (Mail User Agent): The client application used to compose, read, send, and receive emails. Examples: Outlook, Thunderbird, Gmail web interface.
- MTA (Mail Transfer Agent): The server software that transfers email between mail servers using SMTP. Examples: Sendmail, Postfix, Microsoft Exchange.
- MDA (Mail Delivery Agent): Receives email from the MTA and stores it in the user's mailbox. Also called Local Delivery Agent (LDA).
- Mail Server: The computer running the MTA/MDA software. Stores user mailboxes and handles routing.
- DNS (Mail Exchanger records): MX records in DNS tell sending MTAs which server accepts email for a domain (e.g., @gmail.com → Google's mail servers).
- Mailbox: The storage location for a user's received email, held on the mail server or downloaded to the client.

How an E-mail System Works

When you send an email, the following sequence of events occurs:

- 1. You compose and send the email in your MUA (e.g., Gmail). The MUA connects to your outgoing mail server (SMTP server) on port 25, 465, or 587.
- 2. Your SMTP server looks up the recipient domain's MX record in DNS to find the destination mail server.
- 3. Your MTA connects to the recipient's MTA and transfers the email via SMTP.
- 4. The recipient's MTA passes the email to the MDA, which stores it in the user's mailbox.
- 5. When the recipient checks their email, their MUA connects to the incoming server using POP3 (Post Office Protocol) or IMAP (Internet Message Access Protocol) to retrieve the message.

SMTP (Simple Mail Transfer Protocol)

SMTP is the protocol used for sending email between mail servers (MTA to MTA) and from client to server. It operates over TCP port 25 (server-to-server), port 587 (client submission), or port 465 (SMTPS — secure). SMTP is a push protocol — the sender pushes email to the receiving server.

MIME — What It Is

MIME (Multipurpose Internet Mail Extensions) is an extension to the original SMTP email protocol that allows email to carry content beyond plain ASCII text — including attachments, images, audio, video, and HTML.

- Without MIME, email could only carry plain ASCII text.

MIME introduced the ability to attach any type of file and include multipart messages.

MIME Header Fields

- MIME-Version: Specifies the MIME version (typically "1.0").
- Content-Type: Specifies the media type and subtype of the content. E.g., text/html, image/jpeg, application/pdf, multipart/mixed.
- Content-Transfer-Encoding: How the content is encoded for transmission. Values: 7bit, 8bit, base64 (for binary data), quoted-printable (for mostly ASCII with some special chars).
- Content-Disposition: Whether content is inline or an attachment. E.g., attachment; filename="doc.pdf".
- Content-ID: Unique identifier for a MIME part.
- Content-Description: Human-readable description of the content part.

MIME Types

- Text: text/plain, text/html, text/css
- Image: image/jpeg, image/png, image/gif, image/svg+xml
- Audio: audio/mpeg (MP3), audio/wav, audio/ogg
- Video: video/mp4, video/ogg, video/webm
- Application: application/pdf, application/zip, application/json, application/octet-stream (generic binary)
- Multipart: multipart/mixed (email with attachments), multipart/form-data (web form uploads), multipart/alternative (email with both plain text and HTML versions)

Mailer Features & Message Components

- Message Components: Header (From, To, CC, BCC, Subject, Date, Message-ID) and Body (the actual message content, possibly multipart).
- Message Composition: MUAs allow rich-text editing, attachments, signatures, and templates.
- Mailer Features: Address book/contacts, folder organization, filters/rules, search, spam filtering, calendar integration (e.g., meeting invitations), encryption (S/MIME, PGP).

Q6 HTML vs DHTML vs XML. JavaScript — Overview, Advantages & Dev Tools. Server-Side Programming. Notes on Chat Rooms, Newsgroups & Mailing Lists.

Introduction

The World Wide Web has evolved through several markup and scripting technologies — from basic HTML to dynamic DHTML and structured XML — complemented by powerful server-side and client-side scripting.

HTML vs DHTML vs XML

- HTML (HyperText Markup Language): A static markup language for creating web page structure and content. Fixed layout; requires page reload. All browsers can render HTML.
- DHTML (Dynamic HTML): Not a separate language but a combination of HTML + CSS + JavaScript + DOM. Allows web pages to change dynamically without reloading. Elements can appear/disappear, animate, or respond to user events in real time. Example: dropdown menus, image sliders.
- XML (eXtensible Markup Language): A meta-language for defining custom markup languages. Designed to store and transport data, not display it. Tags are user-defined and self-descriptive. Platform and language neutral. Example: RSS feeds, configuration files.

JavaScript as a Scripting Language

JavaScript is the dominant client-side scripting language of the Web.

- Client-side: Runs in the browser, no server interaction required for basic operations. enables real-time interactivity.
- Event-driven: Responds to user actions through event handlers like onclick, onmouseover, onsubmit.
- Interpreted: No compilation needed. The browser's JavaScript engine (V8 in Chrome) executes the code.
- Object-based (not fully OOP): Supports objects and prototype-based inheritance but does not have classical class-based OOP (though ES6+ added class syntax).
- Case-sensitive: myVar and myvar are different identifiers.

Advantages of JavaScript

- Reduces server load by validating forms client-side before submission.
- Creates rich, interactive user interfaces
- Asynchronous programming via AJAX allows updating parts of a page without full reload.
- Massive ecosystem: Node.js enables server-side JavaScript; frameworks like React, Angular, Vue.

JavaScript Development Tools

- Browser DevTools: Chrome DevTools, Firefox Developer Tools — debugging, profiling, DOM inspection, network monitoring. Available by pressing F12.
- Text Editors/IDEs: VS Code, WebStorm, Sublime Text — offer syntax highlighting, auto-complete, and debugging.
- Package Managers: npm (Node Package Manager), yarn — manage JS libraries and dependencies.
- Build Tools: Webpack, Babel — bundle modules and transpile modern JS to older compatible versions.
- Testing Frameworks: Jest, Mocha — automated unit testing.

Server-Side Programming Concepts

Server-side programming refers to scripts that run on the web server in response to client requests. Unlike client-side code which runs in the browser, server-side code runs on the server and generates dynamic HTML that is then sent to the client.

- Technologies: PHP, Node.js, Python (Django/Flask), Java (Servlets/JSP), Ruby on Rails, ASP.NET.
- Tasks: Database queries, authentication/authorization, file I/O, business logic processing, session management, sending emails.
- The client (browser) never sees the server-side source code — only the output HTML is sent.

Chat Rooms

A chat room is an online platform where multiple users can send and receive text messages in real time. Messages appear to all participants simultaneously. Modern chat rooms use WebSockets for real-time bidirectional communication (e.g., Discord, Slack). Early chat rooms used IRC (Internet Relay Chat).

Newsgroups

Newsgroups (Usenet groups) are an early Internet discussion system, predating the Web. They function like bulletin boards where users post messages organized by topic (e.g., comp.lang.java). Messages are distributed across Usenet servers globally using the NNTP protocol. Usenet groups were categorized into hierarchies: comp (computers), sci (science), rec (recreation), soc (society), etc.

Mailing Lists

A mailing list is a collection of email addresses grouped under a single address. When a message is sent to the list address, it is forwarded to all subscribers. Used for announcements (one-way) or discussions (two-way, where members can reply). Software: Mailman, Listserv.

Q7 What is a Web Server? Differentiate Apache, IIS & Microsoft PWS. Apache Configuration Directives & How to Start/Stop Servers.

Introduction

A web server is a software application (and/or hardware) that stores, processes, and serves web pages to clients (browsers) over the HTTP/HTTPS protocol. When a user requests a URL, the web server locates the resource and sends it back as an HTTP response.

- Core functions: Accepting HTTP requests, locating the requested resource (file, script output), returning HTTP responses, managing virtual hosts, logging.
- Popular web servers: Apache HTTP Server, Microsoft IIS, Nginx, LiteSpeed, Microsoft PWS.

Apache HTTP Server

Apache is the most widely deployed open-source web server, developed by the Apache Software Foundation. It runs on Linux, Windows, macOS, and most Unix-like systems. Apache uses a modular architecture — functionality is added via modules (e.g., mod_ssl for HTTPS, mod_rewrite for URL rewriting, mod_php for PHP support).

- Default configuration file: httpd.conf
- Apache uses a process/thread-based model, creating a new process or thread for each connection.

IIS (Internet Information Server)

IIS (Internet Information Services) is Microsoft’s web server, included with Windows Server. It is tightly integrated with the Windows OS and supports ASP.NET, PHP, and classic ASP.

- Configured via the IIS Manager GUI or XML-based configuration files (web.config).

- Supports Windows Authentication, integrated with Active Directory.
- Uses ISAPI (Internet Server Application Programming Interface) for extensions.
- Best suited for Windows-based hosting environments.

Microsoft PWS (Personal Web Server)

PWS was a lightweight version of IIS designed for personal/development use on Windows 95/98/NT Workstation. It allowed developers to test web pages locally without a full server setup.

- Supported basic HTTP, ASP (Active Server Pages), and FrontPage extensions.
- Not suitable for production; replaced by IIS for development on modern Windows with IIS Express.

Apache vs IIS vs PWS — Key Differences

- Apache: Open-source, cross-platform, module-based, highly configurable, large community.
- IIS: Microsoft-proprietary, Windows-only, deeply integrated with Windows/.NET, strong GUI management.
- PWS: Lightweight, for personal/dev use only, Windows 9x era, now obsolete.

Apache Configuration Directives

Apache is configured primarily through the httpd.conf file (and supplementary .conf files in the conf.d/ directory). Key directives include:

- ServerRoot: Base directory for server files. E.g., ServerRoot "/etc/httpd"
- Listen: Port(s) to listen on. E.g., Listen 80 or Listen 443
- DocumentRoot: The directory from which Apache serves files. E.g., DocumentRoot "/var/www/html"
- DirectoryIndex: Default file to serve for a directory request. E.g., DirectoryIndex index.html index.php
- ServerName: The server’s hostname and port. E.g., ServerName www.example.com:80
- <Directory>: Sets permissions and options for a directory. Options Indexes FollowSymLinks; AllowOverride All; Require all granted.
- VirtualHost: Configures virtual hosting to serve multiple websites from one server.
- ErrorLog / CustomLog: Define log file locations and formats.
- LoadModule: Loads additional Apache modules.
- <Location>: Applies directives to a URL path rather than a filesystem path.

Starting and Stopping Apache

On Linux/Unix systems, Apache (httpd or apache2) is managed via system commands:

- Start: sudo systemctl start apache2 (Ubuntu/Debian) OR sudo service httpd start (RHEL/CentOS)
- Stop: sudo systemctl stop apache2
- Restart: sudo systemctl restart apache2 (full restart) OR sudo systemctl reload apache2 (graceful reload without dropping connections)
- Status: sudo systemctl status apache2 — shows if running, uptime, recent logs.
- Enable on boot: sudo systemctl enable apache2.

On Windows, Apache can be started/stopped via the Apache Service Monitor in the system tray, via the Services console (services.msc), or via command line: net start Apache2.4 / net stop Apache2.4.

Q8 Threats & Attacks to Internet Security. Symmetric & Asymmetric Encryption. Digital Signatures. Firewalls (Types). IDS & Why IDS is Preferred Over Firewalls.

Introduction to Internet Security

As reliance on the Internet grows, so do threats to its security. Understanding these threats and the mechanisms used to counter them is critical for protecting data, systems, and users.

Threats & Attacks to Internet/Computer Security

- Malware: Malicious software — viruses, worms, Trojans, ransomware, spyware. Spreads via email attachments, malicious websites, infected drives.
- Phishing: Deceptive emails/websites trick users into revealing sensitive information (passwords, credit card numbers) by impersonating legitimate entities.
- DoS / DDoS: Denial-of-Service / Distributed DoS. Floods a server with so much traffic it becomes unavailable to legitimate users. DDoS uses a botnet.
- Man-in-the-Middle (MITM): Attacker secretly intercepts and possibly alters communication between two parties. Common on unsecured Wi-Fi networks.
- SQL Injection: Inserting malicious SQL code into web form inputs to manipulate or dump a database.
- Cross-Site Scripting (XSS): Injecting malicious scripts into web pages viewed by other users.
- Brute Force: Systematically trying all possible passwords until the correct one is found.
- Zero-Day Exploit: Attack targeting an unknown (unpatched) software vulnerability.

Encryption Schemes

- Symmetric Encryption: Same key is used for both encryption and decryption. Fast and efficient for large amounts of data. Key distribution problem: how to securely share the key. Examples: AES (Advanced Encryption Standard), DES (Data Encryption Standard), 3DES. Use case: Encrypting files, database encryption, disk encryption.
- Asymmetric Encryption (Public Key Cryptography): Uses a key pair — a public key (shared with everyone) and a private key (kept secret). Data encrypted with the public key can only be decrypted with the private key. Slower than symmetric but solves the key distribution problem. Examples: RSA, ECC (Elliptic Curve Cryptography), Diffie-Hellman. Use case: SSL/TLS (HTTPS), email encryption (PGP).
- Hybrid Encryption: Most real-world systems (like TLS/HTTPS) use both — asymmetric encryption to securely exchange a symmetric session key, then symmetric encryption for the actual data transfer. This combines speed and security.

Digital Signatures

A digital signature is a cryptographic mechanism that proves the authenticity and integrity of a digital document.

- How it works: The sender hashes the message (creates a fixed-length digest), then encrypts the hash with their private key. This encrypted hash is the digital signature. The recipient decrypts the signature with the sender’s public key to retrieve the hash, then independently hashes the received message. If the two hashes match, the signature is valid — confirming the message came from the stated sender and has not been tampered with.
- Provides: Authentication (proves sender identity), Integrity (message unchanged), Non-repudiation (sender cannot deny sending).

Firewalls — Types

- Packet Filtering Firewall: Inspects each packet’s header (source/destination IP, port, protocol) against rules. Fast but does not inspect packet payload. Examples: iptables
- Stateful Inspection Firewall: Tracks the state of active connections. More intelligent — knows whether an incoming packet is part of an established connection
- Application/Proxy Firewall: Operates at the application layer. Acts as an intermediary (proxy) between client and server, deeply inspecting HTTP, FTP, DNS traffic.
- Next-Generation Firewall (NGFW): Combines stateful inspection, deep packet inspection (DPI), intrusion prevention, SSL decryption, and identity awareness. Examples: Palo Alto, Fortinet, Cisco Firepower.

IDS (Intrusion Detection System)

An IDS monitors network traffic or host activity for suspicious patterns and generates alerts when potential intrusions or policy violations are detected.

- NIDS (Network-based IDS): Monitors the entire network by analyzing captured packets at strategic points.
- HIDS (Host-based IDS): Monitors a single host — system calls, file integrity, log analysis. Example: OSSEC, Tripwire.
- Signature-based detection: Compares traffic against a database of known attack signatures. Fast and accurate for known threats but misses new/unknown attacks.
- Anomaly-based detection: Establishes a baseline of normal behavior and flags deviations. Can detect new attacks but may have higher false-positive rates.

Why IDS Is Preferred Over Firewalls

Firewalls are perimeter-based — they block or allow traffic based on rules but cannot detect sophisticated attacks that pass through legitimate channels. Reasons IDS is preferred/complementary:

- Insider threats: Firewalls protect only external traffic. IDS detects malicious activity from inside the network (disgruntled employees, compromised internal machines).
- Encrypted traffic: Firewalls may pass encrypted traffic without inspection. IDS (especially with SSL decryption) can analyze content for malicious patterns.
- Post-breach detection: If a firewall is bypassed, IDS can still detect and alert on malicious activity inside the network.
- Forensics: IDS logs detailed attack information useful for forensic analysis and understanding breach scope.

Q1 Classify operating systems and explain different OS structures (Monolithic, Layered, Microkernel, Virtual Machine) with neat diagrams. Explain multiprogramming, batch processing, time-sharing, real-time systems. What is the need for protection and how is it achieved?

Introduction: An Operating System is system software that manages hardware and software resources, acting as an intermediary between users and hardware. Understanding OS types and internal structures is fundamental to computer science.

Types of Operating Systems:

Batch Processing: Jobs with similar requirements are grouped and processed sequentially without user interaction. OS automatically moves from one job to the next. Advantage: high CPU utilisation. Disadvantage: no interactivity, long turnaround time. Example: IBM mainframe payroll processing.

Multiprogramming: Multiple jobs reside in memory simultaneously. When one job waits for I/O, CPU switches to another — CPU is never idle. Significantly increases throughput.

Time-Sharing (Multitasking): Extension of multiprogramming where CPU switches between jobs so rapidly (time quantum) that each user feels they have a dedicated machine. Enables interactive use. Response time typically under 1 second. Example: UNIX, Linux.

Real-Time OS: Processes data within strict time constraints. Hard RTOS: missing deadline = catastrophic failure (aircraft control, pacemakers). Soft RTOS: occasional miss tolerable (multimedia streaming). Example: VxWorks, FreeRTOS.

Distributed OS: Manages networked independent computers as one coherent system. Users access remote resources transparently. Provides fault tolerance and load balancing.

Embedded OS: Designed for resource-constrained devices. Example: Android in smartphones, FreeRTOS in microcontrollers.

OS Structures:

Monolithic Structure: All OS services (file system, device drivers, memory management) packed into one large kernel. No module separation. Advantage: fast — direct function calls. Disadvantage: extremely difficult to maintain; one bug crashes entire system. Example: original UNIX.

Layered Structure: OS divided into layers (0=hardware, N=user interface). Each layer uses only services from the layer directly below. Advantage: modularity, easy debugging. Disadvantage: performance overhead; difficult to strictly define layer boundaries. Example: THE system.

Microkernel Structure: Minimal kernel contains only IPC, basic memory management, and elementary scheduling. All other services (file system, drivers, networking) run as user-space server processes communicating via message passing. Advantage: reliable (server crash doesn't bring down kernel), extensible, portable, more secure. Disadvantage: performance overhead from user-kernel context switches and message passing. Example: Mach (macOS foundation), QNX, MINIX.

Virtual Machine: Hypervisor layer sits between hardware and multiple OS instances, giving each the illusion of dedicated hardware. Type 1 (bare-metal): runs directly on hardware — VMware ESXi, Xen. Type 2 (hosted): runs on host OS — VirtualBox. Advantage: complete isolation between VMs, run multiple OS simultaneously.

Q2 Explain the purpose of system calls with categories and examples (process control, file management, device management, communication). Describe services provided by an OS. Explain storage structure and hierarchy (registers → cache → RAM → disk). Write a note on I/O structure and DMA.

System Calls, OS Services, Storage Hierarchy & DMA

Introduction: An OS provides a rich set of services to users and programs. System calls are the programmatic interface through which user programs request these services from the kernel. The storage hierarchy determines how data moves between CPU and storage. Understanding these mechanisms is central to OS design and performance.

Services Provided by an OS:

User-facing services: Program execution (load and run programs, report errors), I/O operations (users cannot directly control hardware), file system manipulation (create/read/write/delete files), communication between processes (shared memory or message passing), error detection (hardware faults, illegal memory access, arithmetic overflow).

System-efficiency services: Resource allocation (CPU, memory, I/O devices among competing processes), accounting (track usage per user/process for billing and auditing), protection and security (prevent unauthorised access to resources).

System Calls: A system call is the mechanism by which a user-space program requests a kernel service. When called, a trap instruction switches CPU to kernel mode, control transfers to the OS, which performs the service and returns the result. System calls are usually accessed through library APIs (POSIX API in Linux, Win32 in Windows) rather than directly. Example: printf() in C internally calls the write() system call.

Storage Structure and Hierarchy: Computer storage organised by speed, cost, and volatility:

Registers: Inside CPU. Sub-nanosecond access. Hundreds of bytes. Fastest, most expensive.

L1/L2/L3 Cache: On/near CPU chip. 1–40 ns. 32KB–64MB. Caches frequently used data using locality of reference principle.

Main Memory (RAM): ~100 ns. GBs. Volatile — lost on power off. Programs must be in RAM to execute.

SSD: ~0.1 ms. Hundreds of GB. Non-volatile, fast, no moving parts.

HDD: ~10 ms. TBs. Non-volatile. Cheapest per GB. Mechanical — slowest.

Optical/Tape: Slowest, cheapest. Used for long-term archival backup.

The principle of locality (temporal: recently used data likely used again; spatial: nearby data likely accessed) makes caching highly effective.

I/O Structure:

Programmed I/O (Polling): CPU continuously checks device status register to see if operation complete. Wastes CPU cycles — CPU spins doing nothing. Suitable only for very fast, brief operations.

Interrupt-Driven I/O: CPU initiates I/O and continues executing other processes. When device completes, it sends a hardware interrupt. CPU suspends current execution, saves state, executes the Interrupt Service Routine (ISR), then resumes. ISR is located through the interrupt vector table (array of ISR addresses indexed by interrupt number). Much more efficient than polling.

DMA (Direct Memory Access): For large bulk transfers (disk reads, network), the DMA controller handles data movement directly between device and main memory without CPU involvement. CPU programs DMA controller with: source address, destination address in RAM, byte count, and direction. DMA autonomously transfers data, cycle-stealing bus access when CPU doesn't need it. On completion, DMA sends ONE interrupt to CPU. CPU is completely free during the entire transfer. Example: Reading a 10MB file — DMA moves all 10MB from disk buffer to RAM while CPU executes other processes; only one interrupt at the end.

Q3 Define deadlock. State and explain the four necessary conditions (Mutual Exclusion, Hold & Wait, No Preemption, Circular Wait). Differentiate deadlock prevention vs avoidance vs detection. Explain Banker's Algorithm for deadlock avoidance with a complete numerical example. How does a system recover from deadlock?

Deadlock: Conditions, Prevention, Avoidance & Banker's Algorithm

Introduction: Deadlock is a situation where a set of processes are permanently blocked because each holds a resource and waits for another resource held by another process in the set — a circular dependency from which no process can proceed. It is a critical OS problem that can halt an entire system.

Four Necessary Conditions (Coffman Conditions): All four must hold simultaneously for deadlock:

1. *Mutual Exclusion:* At least one resource held in non-shareable mode — only one process can use it at a time. Others must wait.
2. *Hold and Wait:* A process holds at least one resource while waiting for additional resources held by other processes.
3. *No Preemption:* Resources cannot be forcibly taken from a process. Released only voluntarily after the process finishes using them.
4. *Circular Wait:* A set of processes {P0, P1, ..., Pn} exists such that P0 waits for P1's resource, P1 waits for P2's, ..., Pn waits for P0's — forming a cycle.

Deadlock Prevention: Eliminate at least one of the four conditions:

Eliminate Hold and Wait: Require processes to request all resources at once before starting. If any unavailable, request nothing. Disadvantage: low resource utilisation, possible starvation.

Allow Preemption: If a process holding resources requests an unavailable resource, preempt all its held resources. Only works for resources whose state can be saved/restored (CPU, memory).

Eliminate Circular Wait: Impose total ordering on all resource types. Processes must always request resources in increasing order. Example: number resources 1–N; always request in ascending order.

Deadlock Avoidance — Banker's Algorithm: Requires each process to declare maximum resource needs in advance. OS dynamically evaluates each request to ensure system stays in a safe state.

Safe State: A state is safe if there exists a safe sequence — an ordering of all processes such that each process's remaining needs can be satisfied using currently available resources plus resources held by all preceding processes in the sequence. Safe state → no deadlock. Unsafe state → possible deadlock.

Data structures (n processes, m resource types):

- Available[m]: currently available instances of each resource type.
- Max[n][m]: maximum demand of each process.
- Allocation[n][m]: currently allocated to each process.
- Need[n][m] = Max – Allocation: remaining needs.

Safety Algorithm:

1. Work = Available; Finish[i] = false for all i.
2. Find i where Finish[i]=false AND Need[i] ≤ Work.
3. If found: Work += Allocation[i], Finish[i] = true. Repeat step 2.
4. If all Finish[i]=true → safe state.

Resource Request Algorithm: When Pi requests Request[i]:

1. If Request[i] > Need[i] → error.
2. If Request[i] > Available → wait.
3. Tentatively: Available -= Request[i], Allocation[i] += Request[i], Need[i] -= Request[i].
4. Run safety algorithm. If safe → grant. If unsafe → rollback and wait.

Q6 Explain paging for memory management — logical vs physical address, page table, TLB. Differentiate internal vs external fragmentation. Explain segmentation and combined paged segmentation.

Paging, Page Replacement & Segmentation

Introduction: Memory management is a critical OS function that must allocate memory to multiple processes, provide protection, and give each process the illusion of a large address space. Paging is the dominant mechanism, extended by virtual memory and page replacement algorithms when physical memory is insufficient.

Fragmentation: *Internal Fragmentation:* Wasted space inside an allocated memory block. Example: process needs 5.5KB, gets 8KB (2 pages of 4KB) — 2.5KB wasted inside the second page. *External Fragmentation:* Total free memory is sufficient but scattered in non-contiguous small holes that cannot satisfy a large contiguous request. Paging eliminates external fragmentation.

Paging: Physical memory divided into fixed-size frames. Logical memory divided into pages of the same size (typically 4KB). A page table per process maps page numbers to frame numbers.

Address Translation: Logical address split into page number (p) and page offset (d). Page table gives frame number (f). Physical address = f × page_size + d.

TLB (Translation Lookaside Buffer): Hardware cache for recent page→frame mappings. ~98% hit rate → nearly instant translation. On miss, access page table in RAM. Effective Access Time = hitrate × TLB_time + (1–hitrate) × (TLB_time + memory_time).

Virtual Memory and Demand Paging: Allows a process to run even when not all its pages are in memory. Pages loaded on demand. On accessing a page not in memory → page fault. OS handles: save state → find free frame → load page from disk → update page table → restart instruction.

Page Replacement Algorithms: When all frames are full and a new page is needed, select a victim page to evict. *FIFO:* Replace the oldest page (longest in memory). Simple — use a queue. Belady’s Anomaly: more frames may cause more faults.

Optimal (OPT): Replace page not used for longest time in future. Minimum possible faults — optimal benchmark. Not implementable in practice (needs future knowledge). *LRU (Least Recently Used):* Replace page not used for longest time in past. Good approximation of Optimal.

Implementation: maintain timestamps or use a stack (most recent at top, LRU at bottom).

Segmentation: Segmentation divides a process into logical segments of variable size (main program, function, stack, heap, symbol table). Logical address = (segment number, offset). Segment table: each entry has base (physical start address) and limit (segment length). Access protection: if offset ≥ limit → segmentation fault.

Advantage: matches program’s logical structure — programmer-friendly. Disadvantage: variable-size segments cause external fragmentation (same problem as variable-partition allocation).

Paged Segmentation (Combined): Each segment is divided into fixed-size pages. Eliminates external fragmentation of pure segmentation while preserving logical segment structure. Address translation: segment number → segment table → page table for that segment → frame number + offset. Used by Intel x86 architecture. Advantage: gets best of both — logical structure of segments, efficiency of paging.

Q7 Explain disk scheduling algorithms: FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK. Apply them on a given request sequence with head starting at a given track. Calculate total head movement and determine which algorithm is most efficient. Write a note on DMA and spooling.

Disk Scheduling Algorithms with Calculations

Introduction: Hard disk drives store data on rotating magnetic platters in concentric tracks divided into sectors. The disk head must physically move (seek) to the correct track before reading or writing. Seek time (5–15ms) dominates disk access time. Since many processes have pending disk I/O, the OS schedules disk requests to minimise total head movement, maximising throughput and minimising response time.

Disk Access Time = Seek Time + Rotational Latency + Transfer Time. Seek time is dominant and directly controllable by the OS scheduling algorithm.

Given Example (2023 paper): 200 tracks (0–199). Requests: {82, 170, 43, 140, 24, 16, 190}. Head starts at track 50.

1. FCFS (First Come First Served): Serve requests in arrival order regardless of head position.

tracks
Simple and fair but highly inefficient — ignores spatial locality of requests.

2. SSTF (Shortest Seek Time First): Always service the request closest to current head position. Greedy approach. Much more efficient than FCFS. Problem: Starvation — requests at extremes may wait indefinitely as closer requests keep arriving.

3. SCAN (Elevator Algorithm): Head moves in one direction servicing all requests, reaches the last request in that direction, then reverses. Like an elevator. Eliminates starvation. Maximum wait time = time for two full sweeps of disk.

4. C-SCAN (Circular SCAN): Like SCAN but after reaching one end, head jumps back to the other end without servicing requests on return. Provides more uniform wait time.

5. LOOK and C-LOOK: LOOK: Like SCAN but head only moves as far as the last request in each direction — doesn’t travel to physical disk end. More efficient than SCAN. C-LOOK: Like C-SCAN but jumps from last request at high end directly to first request at low end (not to track 0). More efficient than C-SCAN.

Comparison Table:

Algorithm	Total Movement	Notes
FCFS	642 tracks	Simple, inefficient
SSTF	208 tracks	Best movement, starvation risk
SCAN	314 tracks	Fair, no starvation
C-SCAN	373 tracks	Most uniform wait times

DMA (Direct Memory Access): DMA allows I/O devices to transfer data directly to/from main memory without CPU involvement. CPU programs DMA controller with: source (device buffer), destination (RAM address), byte count, and direction. DMA controller autonomously transfers data, cycle-stealing bus access when CPU doesn’t need it. On completion, DMA sends ONE interrupt to CPU. CPU is completely free during entire transfer. Example: Reading 1MB file from disk — DMA transfers all 1MB to RAM while CPU runs other processes; only one interrupt at the end. Dramatically reduces CPU overhead for bulk I/O.

Spooling (Simultaneous Peripheral Operations On-Line): Output destined for a slow device (printer) is first written to a fast disk buffer (the spool). OS daemon reads from spool and feeds the slow device at its pace. Applications don’t block waiting for the slow device — they write to spool and continue immediately. Example: Printing — document data goes to print spool disk buffer instantly, application returns to user control, print spooler daemon feeds data to printer in the background.

Q8 Discuss file allocation methods in detail: Contiguous, Linked, and Indexed allocation — with advantages and disadvantages of each. Explain directory structures: single-level, two-level, tree-structured. Explain file system protection, access control lists, and the UNIX file system.

File Allocation Methods, Directory Structures & UNIX File System

Introduction: A file system is the OS component responsible for organising, storing, retrieving, protecting, and naming files on persistent storage. Two fundamental design questions are: how to allocate disk space to files, and how to organise the file namespace. This answer covers all three allocation methods, directory structures, and the UNIX file system.

File Allocation Methods:

1. Contiguous Allocation: Each file occupies a set of adjacent disk blocks. Directory entry stores only starting block number and file length. Example: file starting at block 14 with length 4 occupies blocks 14, 15, 16, 17.

Advantages: Simple — only start and length needed. Excellent read performance — sequential or random access both fast. Supports direct access: to read block i, access block (start+i).

Disadvantages: External fragmentation — holes scattered across disk over time. File cannot grow without moving elsewhere. Must declare file size at creation time.

2. Linked Allocation: Each file is a linked list of disk blocks anywhere on disk. Each block contains a pointer to the next block. Directory stores first and last block addresses.

Example: file starts at block 9 → 16 → 1 → 10 → 25 → NULL. Advantages: No external fragmentation — any free block usable. Files grow dynamically — just allocate any free block.

Disadvantages: Sequential access only — random access requires following pointers from start O(n). Pointer overhead wastes space inside blocks. Poor reliability — corrupt pointer loses all subsequent data.

FAT (File Allocation Table): Brings all pointers into a table in memory. FAT[i] = next block number (or EOF/free). Faster traversal since FAT cached in RAM. Used by USB drives, SD cards.

3. Indexed Allocation: Each file has a dedicated index block containing an array of pointers to all its data blocks. Directory stores only the index block address.

Advantages: Direct access O(1) — read index_block[i] to get block i. No external fragmentation. No pointer overhead in data blocks.

Disadvantages: Index block overhead — even tiny files need a full index block. Single index block insufficient for very large files.

Multi-level Indexing (UNIX inode): 12 direct pointers + 1 single indirect (points to block of pointers) + 1 double indirect + 1 triple indirect. Supports files from tiny to hundreds of GB.

Comparison:

Method	External Fragmentation	Direct Access	Grow File	Reliability
Contiguous	Yes	Excellent	Difficult	Good
Linked	No	Poor O(n)	Easy	Poor
Indexed	No	Good O(1)	Easy	Good